Software Engineering Lab

Building .NET Applications on the Oracle Database with Microsoft Visual Studio

Lab 5 – Disconnected Layer

Agenda

Disconnected Mode

- OracleDataAdapter
- DataSets
- OracleCommandBuilder

Disconnected Mode

- This mode allows you to load data from server, disconnect from the server and work on the loaded data then update the server later.

- It is the **OracleDataAdapter** that manages connections with the data source and gives us ***disconnected*** behavior.

Disconnected Mode

- **A DataSet** is an in-memory data store that can hold a collection of tables each represented by DataTable object.

- **An OracleDataAdapter** uses Dataset to **fill** each **DataTable** in a DataSet with data from the Oracle database. The Dataset contains the copy of the data we requested

- An **OracleDataAdapter updates** changes in the DataSet to the Oracle database.

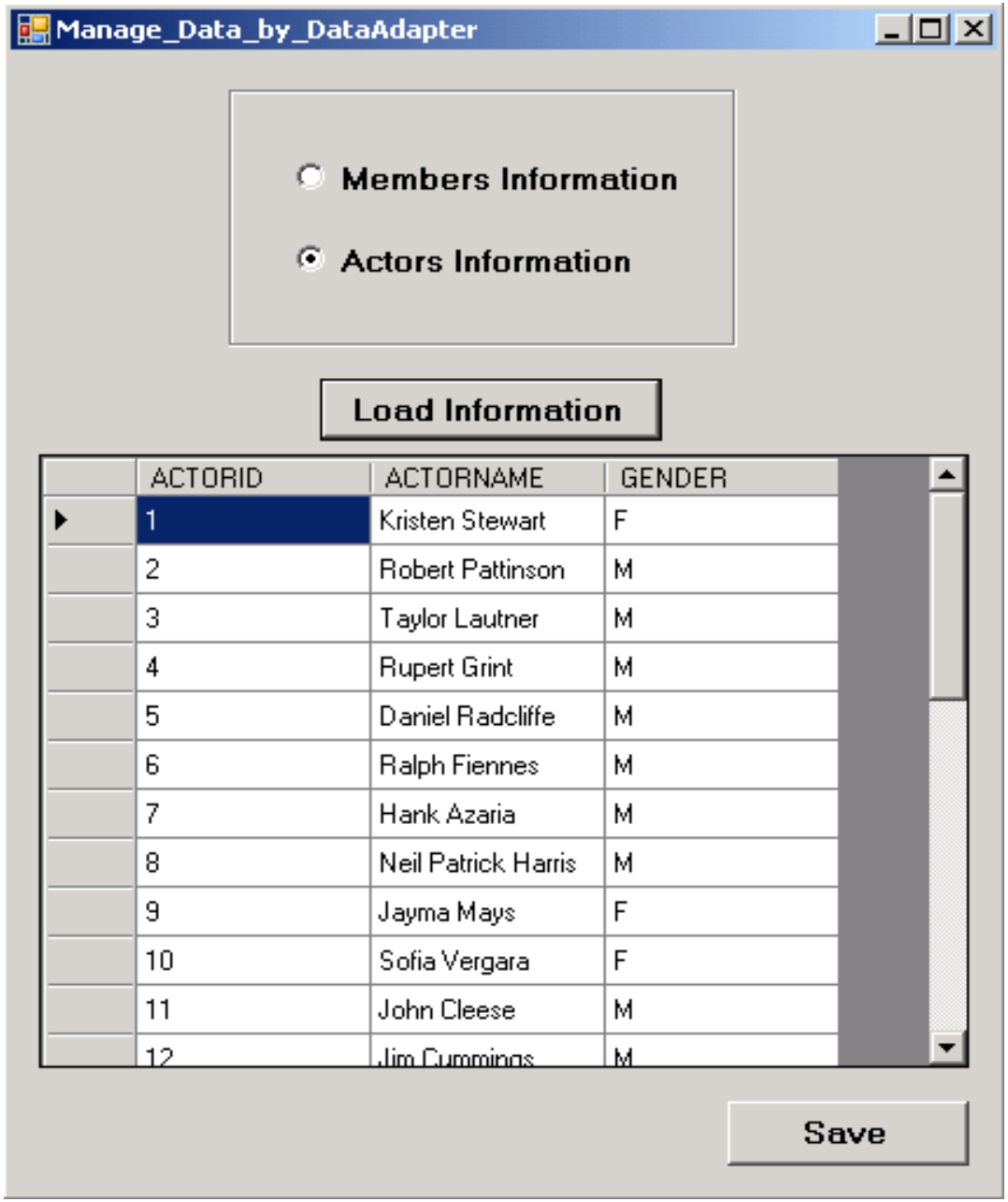
Disconnected Mode

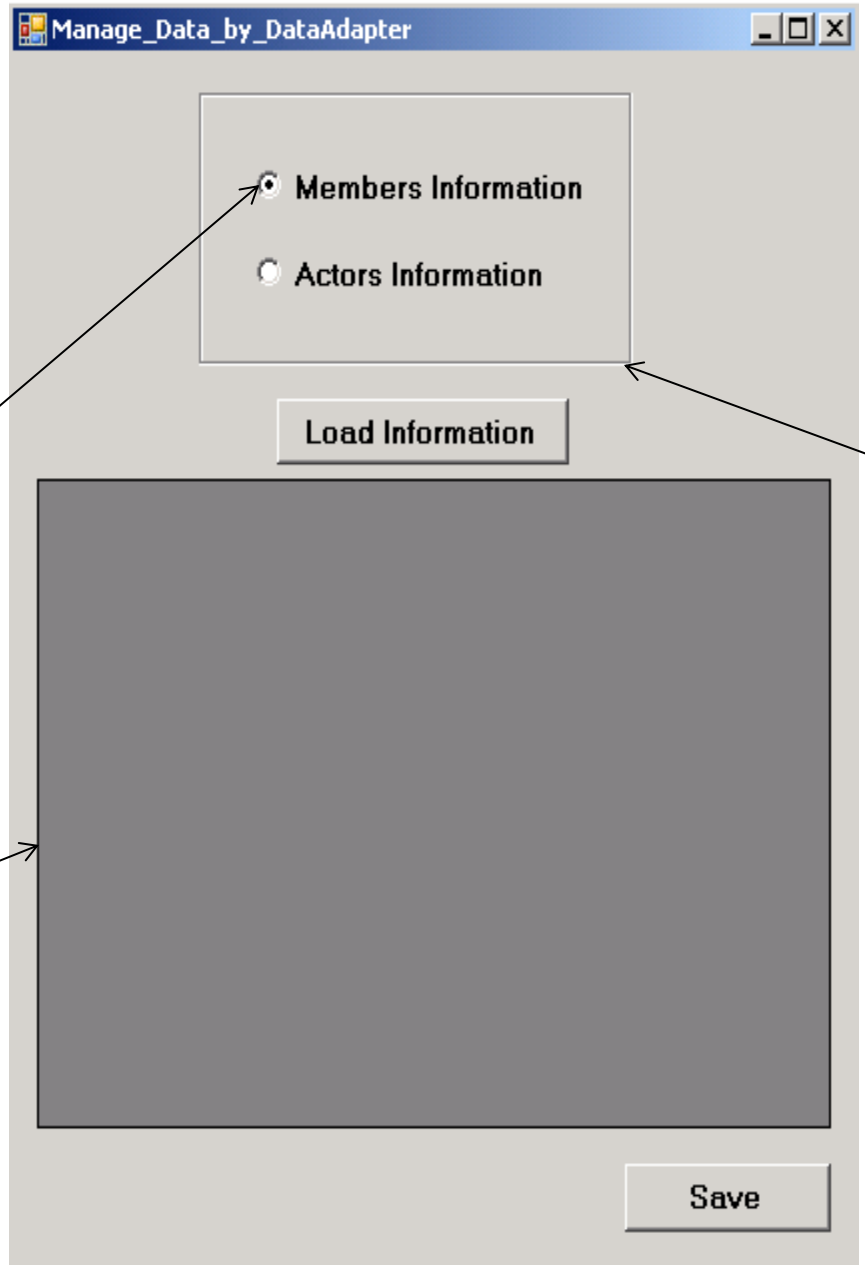
- The **OracleDataAdapter** performs the following tasks when *filling* a **DataSet** with data:
  - Open connection
  - Retrieve data into DataSet
  - Close connection

Using Disconnected Mode

1. Create Dataset object
2. Create OracleDataAdapter object
   1. Command string
   2. Connection string
3. Add command parameters (if any)
4. Call dataAdapter.Fill(dataset).
5. Display from dataset on DataGridView

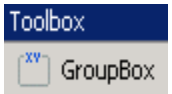
Manage Members or Actors information using Oracle Data Adapter





RadioButton

GroupBox



DataGridView

```
using System.Windows.Forms;
using Oracle.DataAccess.Client;
using Oracle.DataAccess.Types;

namespace DisconnectedMode_Startup
{
    public partial class Manage_Data_DataAdapter : Form
    {
        OracleDataAdapter adapter;
        OracleCommandBuilder builder;
        DataSet ds;
```

In Load_Information_Button, Detect which radio button is selected

```
private void loadInformation_Click(object sender, EventArgs e)
{
    string constr = "Data Source=orcl; User Id=scott;Password=tiger;";

    string cmdstr = "";

    if (rdo_members.Checked)
        cmdstr = "select * from members";

    else if (rdo_actors.Checked)
        cmdstr = "select * from actors";
```

- *Create a Data Adapter Object:*

```
adapter=new OracleDataAdapter(cmdstr, constr);
```

- *Create a DataSet Object:*

```
DataSet ds = new DataSet();
```

- *Filling the Dataset with the data*

```
adapter.Fill(ds);
```

- *Populate data grid control with this data table*

```
dataGridView1.DataSource = ds.Tables[0];
```

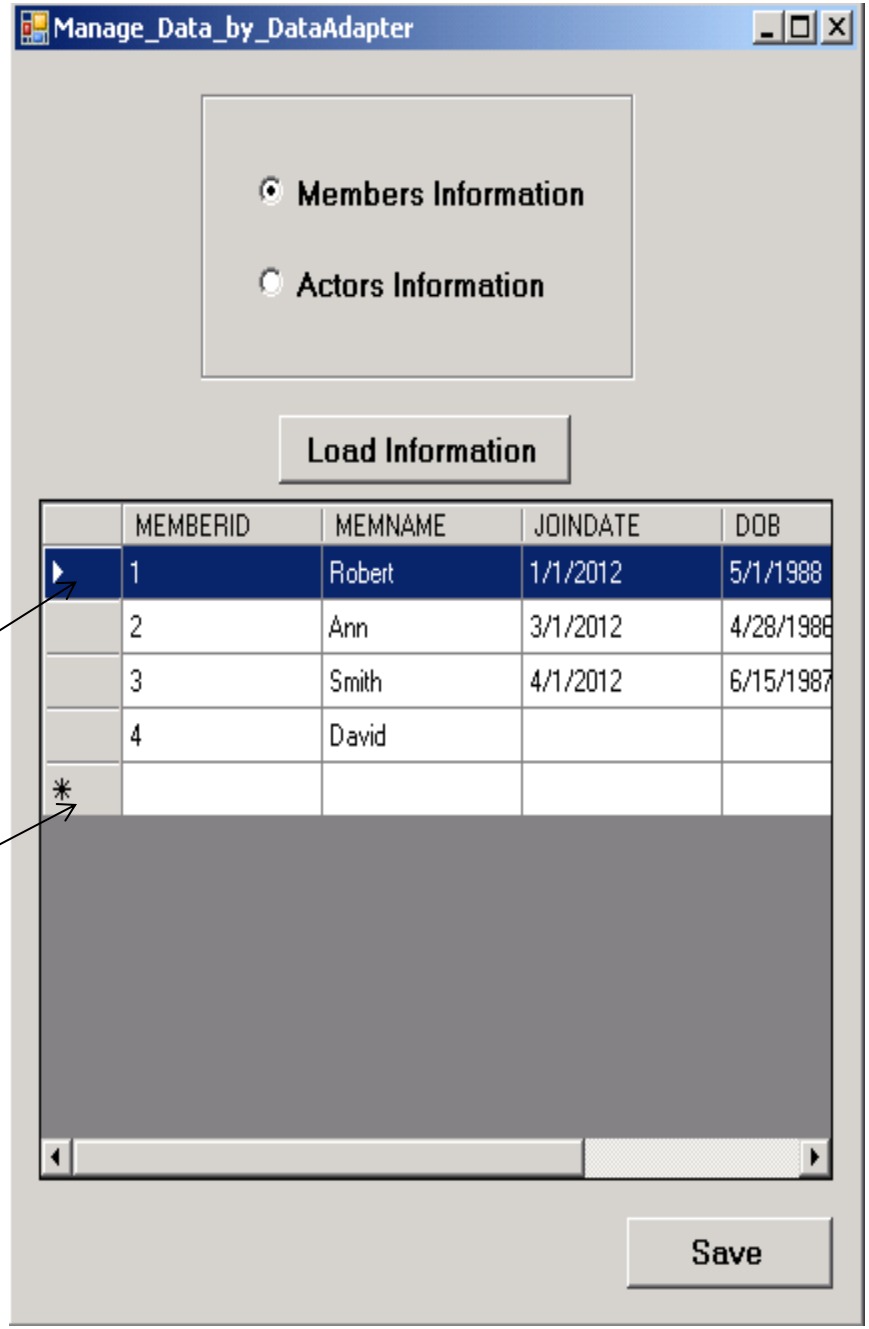
```
private void loadInformation_Click(object sender, EventArgs e)
{
    string constr = "Data Source=orcl; User Id=scott;Password=tiger;";

    string cmdstr = "";

    if (rdo_members.Checked)
        cmdstr = "select * from members";
    else if (rdo_actors.Checked)
        cmdstr = "select * from actors";

    adapter = new OracleDataAdapter(cmdstr, constr);
    ds = new DataSet();
    adapter.Fill(ds);
    dataGridView1.DataSource = ds.Tables[0];
}
```

How To Save changes (Insert, Update & Delete)?



Delete or Update

Insert

*In **btn_Save** Event Handler:*

- Create **OracleCommandBuilder** object to provide automatic SQL generation for the **OracleDataAdapter** when updates are made to the database.
- Create the builder for the adapter to automatically generate the Command when needed.

```
OracleCommandBuilder builder =
        new OracleCommandBuilder(adapter);
```

*In **btn_Save** Event Handler:*

- The OracleCommandBuilder will create the UpdateCommand for the adapter to update the required table

```
adapter.Update(ds.Tables[0]);
```

```
private void save_Click(object sender, EventArgs e)
{
    builder = new OracleCommandBuilder(adapter);
    adapter.Update(ds.Tables[0]);
}
```

Thank You